PRIME SCHOOL

Science

Year 8

Student Manual

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Before you begin

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Unit 1 also includes a Final project, Check your progress, Evaluation, What can you do, and a Key words round-up.

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INTRODUCTION

How this book works

Science is not a list of facts to memorise. It is a way of asking questions and finding answers you can trust. In this first unit you will follow oxygen from the air outside into a mitochondrion inside a muscle cell, and discover that breathing and respiration are not the same thing at all.

What makes this book different

- **Everything is original.** Every sentence, example and diagram was made for

this book.

- **Every claim is checked.** Facts come from real, cited sources; every

numerical answer was executed by computer before publication.

- **Thinking is taught explicitly.** Dedicated panels show you how specialists

in this subject actually reason.

A note to teachers and parents

This book follows the spirit of an international lower-secondary programme, interpreted afresh for Prime School. Each topic opens with clear outcomes and closes with self-assessment, so progress is visible to pupil and teacher alike.

HOW TO USE THIS BOOK

Reading the page

Each section follows the same rhythm, so you always know where you are.

PANEL	WHAT IT DOES
Learning outcomes	What you will be able to do by the end.
Getting started	A warm-up that wakes up what you know.
Key words	The vocabulary of the topic, defined.
Worked example	A model solution, with the reasoning shown.
Questions	Practice, from straightforward to demanding.
Investigation	A hands-on experiment with equipment.
Safety	How to carry out the work safely.
Think like a scientist	A task that builds scientific thinking.
Summary checklist	'I can' statements to check your progress.

How to work

Read with a pencil in your hand. When the book asks a question, try to answer it before reading on. Do the work in your notebook, and check your reasoning, not just your answer.

GETTING SET UP

Before Unit 1

You need a pencil, a ruler, a notebook and your curiosity. Anything else this unit requires is listed where it is needed.

Your toolkit

- A notebook for working, planning and recording, not just for final answers.
- The habit of checking: recalculate, reread, and ask whether the answer is sensible.
- The confidence to say "I am not sure yet", which is where all real learning starts.

A quick self-check

1. What do you already know about this topic?
2. Which part of it do you expect to find hardest, and why?
3. How will you know when you have genuinely understood something, rather than just recognised it?

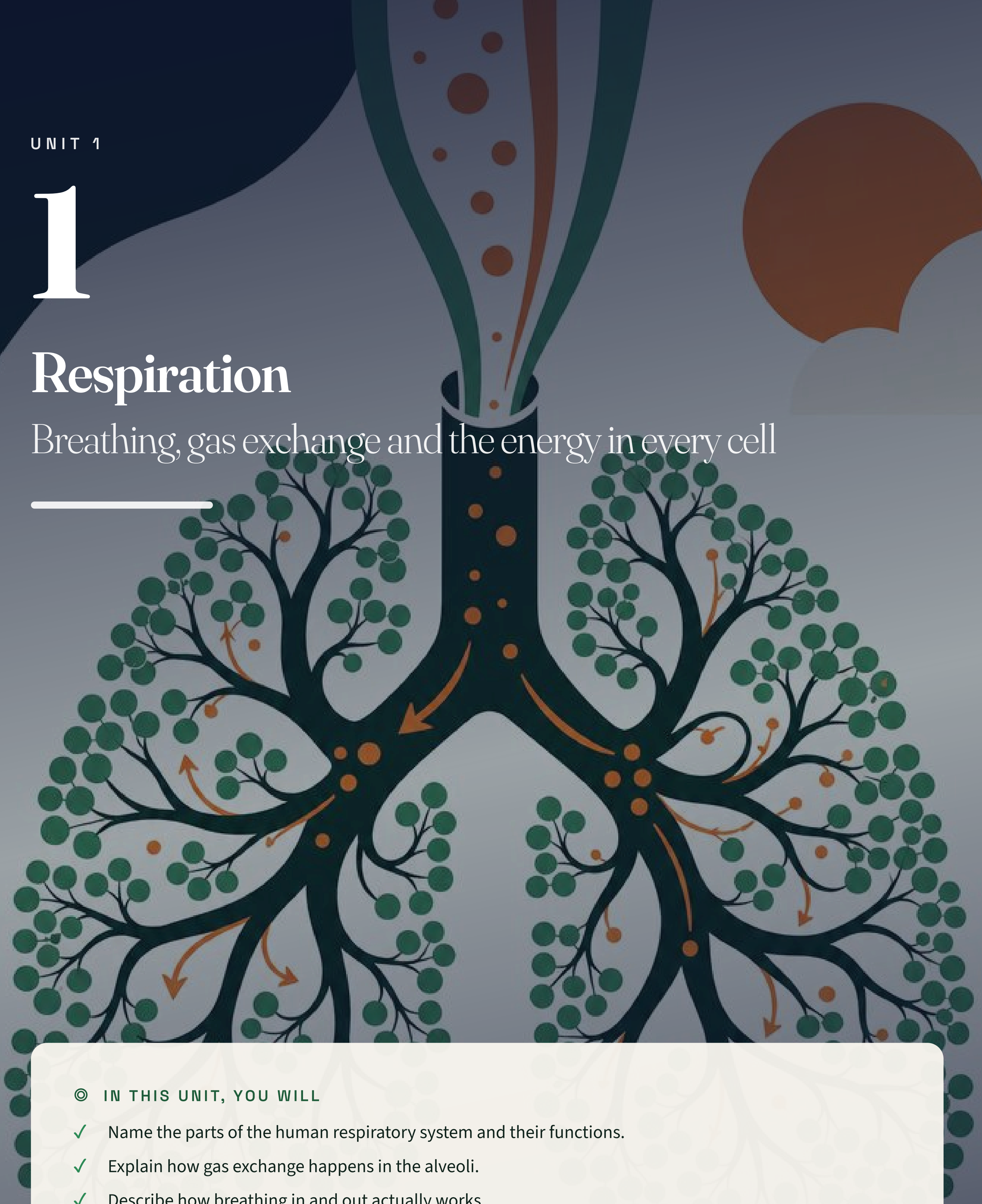
Unit 1 begins now. Work carefully, check honestly, and ask good questions.

UNIT 1

1

Respiration

Breathing, gas exchange and the energy in every cell



IN THIS UNIT, YOU WILL

- ✓ Name the parts of the human respiratory system and their functions.
- ✓ Explain how gas exchange happens in the alveoli.
- ✓ Describe how breathing in and out actually works.
- ✓ Distinguish breathing from respiration.
- ✓ Explain how blood carries oxygen around the body.

✦ GETTING STARTED

Sit still and count how many breaths you take in one minute. Compare with a partner. Now run on the spot for thirty seconds and count again. Discuss:

- why did the number change?
- what do you think your body needed more of?
- where in your body do you think that thing was going?

UNIT 1 · TOPIC 1.1

The human respiratory system

Learn

Every cell in your body needs energy, and to release that energy it needs **oxygen**. Your **respiratory system** is the set of organs that gets oxygen from the air into your blood, and gets **carbon dioxide** out again.

Air travels a fixed route. It enters through your nose or mouth, passes down the **trachea**, which splits into two **bronchi**, one for each lung. Inside the lungs, each bronchus branches into thousands of narrower **bronchioles**, which end in tiny air sacs called **alveoli**.

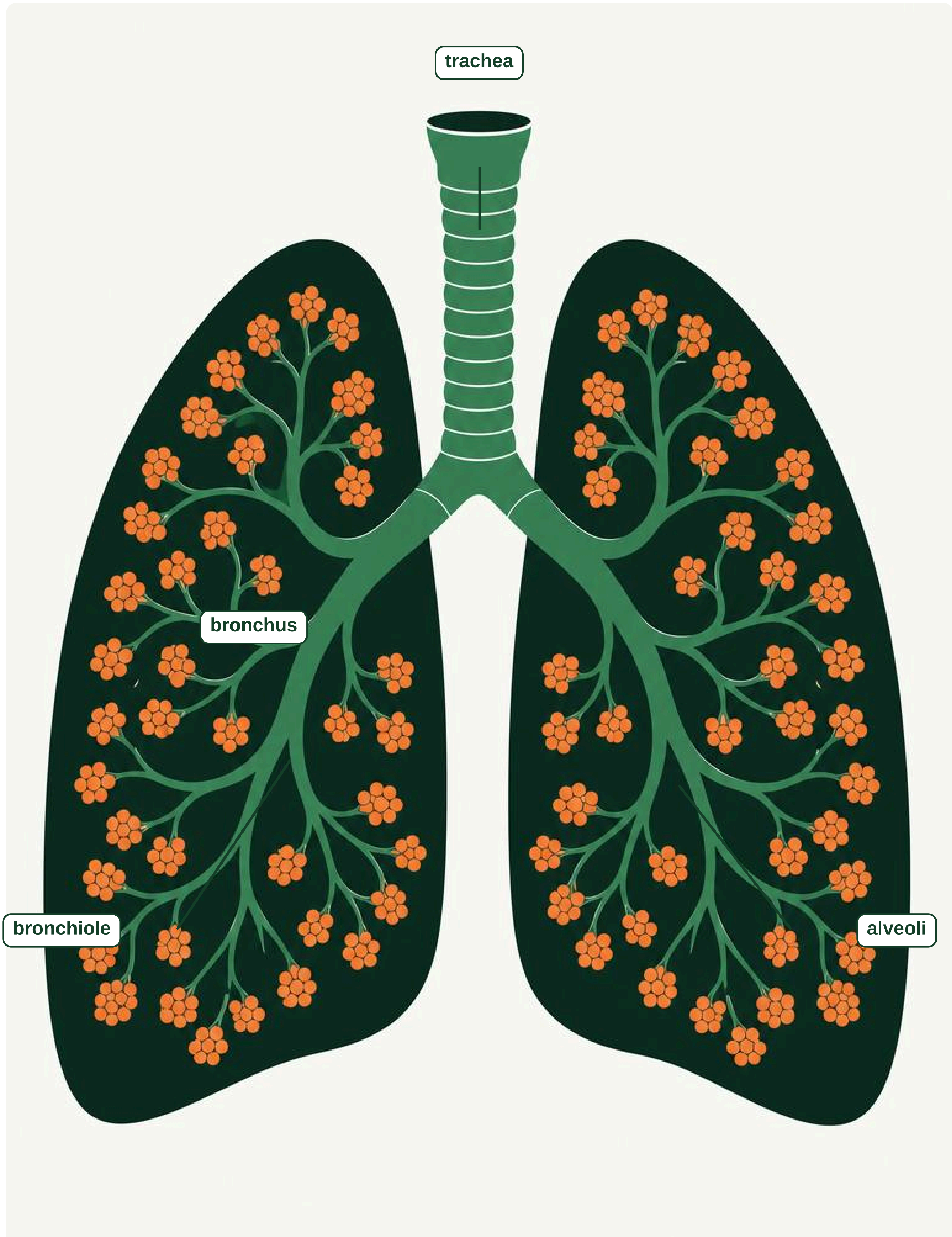


Figure 1.1 The human respiratory system, with the main parts labelled.

PART	WHAT IT DOES
trachea	The windpipe. Carries air from the throat towards the lungs. Rings of cartilage hold it open.
bronchus	One of two tubes carrying air into each lung. (Plural: bronchi.)
bronchiole	A narrow branch of a bronchus, deep inside the lung.
alveolus	A tiny air sac where gases pass into and out of the blood. (Plural: alveoli.)
diaphragm	A sheet of muscle below the lungs that makes you breathe.
ribs	Bones that protect the lungs and help move air.

The lining of your airways is covered in tiny hairs called **cilia**, and in sticky **mucus**. Dust and germs get trapped in the mucus, and the cilia sweep it up and away from your lungs. This is why smoking is so damaging: it destroys the cilia, so the lungs cannot clean themselves.

QUESTIONS

1. Put these in the order that air passes through them: bronchiole, trachea, alveolus, bronchus.
2. Why does the trachea need rings of cartilage?
3. Explain how cilia and mucus protect the lungs.
4. Suggest why a smoker often has a cough.

✧ DID YOU KNOW?

Your lungs contain roughly 300 to 500 million alveoli. Spread flat, their total surface would cover an area about the size of a tennis court, all folded neatly inside your chest.

KEY WORDS

- **respiration**: the release of energy from food inside cells.
- **trachea**: the windpipe, carrying air towards the lungs.
- **bronchus**: a tube carrying air into a lung.
- **bronchiole**: a narrow air tube inside the lung.
- **alveolus**: a tiny air sac where gas exchange happens.
- **cilia**: tiny hairs that sweep mucus out of the airways.

UNIT 1 · TOPIC 1.2

Gas exchange

Learn

Gas exchange is the swap that happens in the alveoli: oxygen moves from the air into the blood, and carbon dioxide moves from the blood into the air.

Gases move by **diffusion**, from where there is more of them to where there is less. The air you breathe in has plenty of oxygen and very little carbon dioxide. The blood arriving at the alveoli is the opposite way round. So oxygen diffuses into the blood, and carbon dioxide diffuses out.

The alveoli are beautifully built for this job:

- there are **millions** of them, giving an enormous surface area
- their walls are only **one cell thick**, so the distance is tiny
- they are wrapped in **capillaries**, so blood is always right there
- they are **moist**, because gases dissolve before they cross

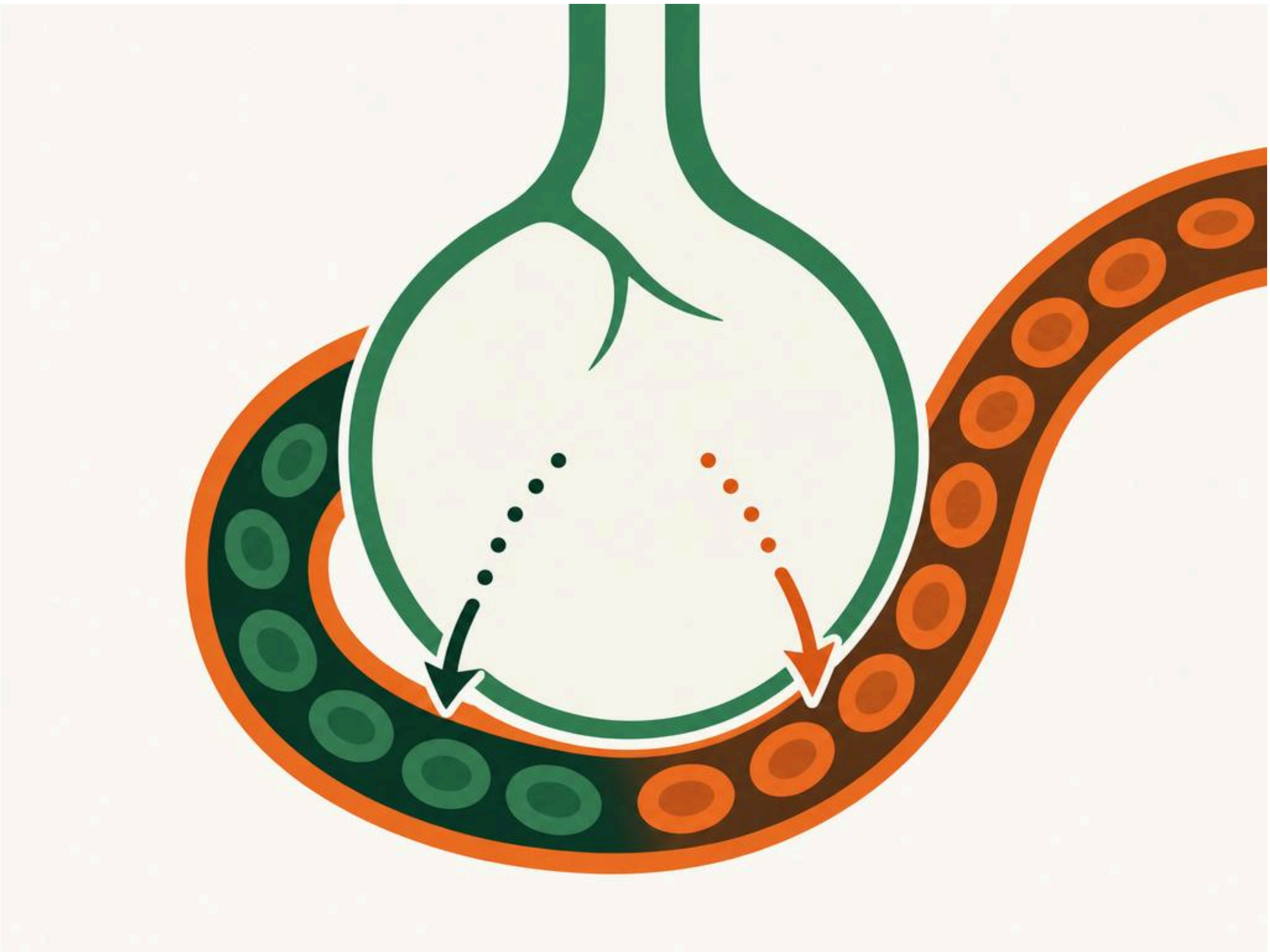


Figure 1.2 An alveolus and its capillary, where gases are exchanged.

GAS	BREATHED IN	BREATHED OUT
oxygen	about 21%	about 16%
carbon dioxide	about 0.04%	about 5%
nitrogen	about 78%	about 78%

QUESTIONS

1. What is gas exchange?
2. Explain what diffusion means in your own words.
3. Give three features of an alveolus that make it good at gas exchange, and say why each one helps.
4. Look at the table. Why does the amount of nitrogen not change?
5. The air you breathe out has 16% oxygen, not 0%. What does this tell you?

∴ THINK LIKE A SCIENTIST

A student says: "We breathe in oxygen and breathe out carbon dioxide." Using the table, explain why this statement is not quite right, and rewrite it so it is accurate.

UNIT 1 · TOPIC 1.3

Breathing

Learn

Breathing is the movement of air into and out of your lungs. It is done by muscles: your **diaphragm** and the muscles between your ribs.

Breathing in (inhaling): the diaphragm contracts and flattens, and the rib muscles pull the ribs up and out. The space inside your chest gets bigger, the pressure inside drops, and air rushes in.

Breathing out (exhaling): the diaphragm relaxes and domes upwards, and the ribs move down and in. The space gets smaller, the pressure rises, and air is pushed out.

Your lungs do not have muscles of their own. They are moved entirely by the muscles around them.

INVESTIGATION: MEASURING BREATHING RATE

You will need: a stopwatch or clock with a second hand, and a partner.

Method:

1. Sit still for two minutes, then count your partner's breaths for one full minute. Record the number.
2. Have your partner step on the spot for one minute, then immediately count their breaths for one minute again. Record it.
3. Wait two minutes, then measure once more, to see how quickly they recover.
4. Repeat with roles swapped, and record all results in a table.

A resting adult usually breathes between 12 and 20 times a minute. Exercise raises this because muscles need more oxygen and produce more carbon dioxide.

⚠ SAFETY

Stop at once if anyone feels dizzy or unwell. Anyone with asthma or a breathing condition should take the role of timer and recorder rather than exercising.

QUESTIONS

1. Describe what your diaphragm does when you breathe in.
2. Explain why air moves into your lungs, in terms of pressure.
3. Why do your lungs need muscles around them?
4. In your investigation, why did you wait two minutes before the first measurement?

UNIT 1 · TOPIC 1.4

Respiration

Learn

Here is the idea that gives this unit its name, and it is not breathing.

Respiration is a chemical reaction inside every one of your cells. It releases the energy stored in food, using oxygen. Its word equation is:

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glucose + oxygen  →  carbon dioxide + water + ENERGY
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Breathing is the movement of air. Respiration is the release of energy inside cells. Breathing supplies respiration with what it needs, and carries away what it makes.

Respiration happens in a part of the cell called the **mitochondrion**. Cells that need lots of energy, such as muscle cells, contain many mitochondria.

QUESTIONS

1. Write the word equation for respiration.
2. Explain the difference between breathing and respiration.
3. Where in a cell does respiration happen?
4. Suggest why a muscle cell has more mitochondria than a skin cell.

KEY WORDS

- **breathing**: the movement of air into and out of the lungs.
- **diffusion**: movement of a substance from a higher to a lower concentration.
- **gas exchange**: the swap of oxygen and carbon dioxide in the alveoli.
- **glucose**: the sugar used in respiration.
- **mitochondrion**: the part of a cell where respiration happens.

UNIT 1 · TOPIC 1.5

Blood

Learn

Oxygen has to be carried from your lungs to every cell, and your **blood** does it. Blood has four main parts.

PART	WHAT IT DOES
red blood cells	Carry oxygen. They contain **haemoglobin**, which binds oxygen in the lungs and releases it in the tissues.
white blood cells	Defend the body against germs.
platelets	Help the blood to clot when you are cut.
plasma	The pale liquid that carries the cells, dissolved carbon dioxide and nutrients.

A red blood cell is beautifully adapted: it has no nucleus, leaving more room for haemoglobin, and it is dimpled on both sides, which increases its surface area for absorbing oxygen.

QUESTIONS

1. Which part of the blood carries oxygen?
2. What is haemoglobin, and what does it do?
3. Give two ways a red blood cell is adapted to carrying oxygen.
4. Which part of the blood would increase if you were fighting an infection?

☑ SUMMARY CHECKLIST

- ✓ I can name the parts of the respiratory system and their functions.
- ✓ I can explain gas exchange and why alveoli are good at it.
- ✓ I can describe how the diaphragm and ribs make me breathe.
- ✓ I can explain the difference between breathing and respiration.
- ✓ I can describe how blood carries oxygen.

UNIT 1 · FINAL PROJECT

The oxygen journey

★ FINAL PROJECT

The oxygen journey

Trace a single oxygen molecule from the air outside a window to a mitochondrion inside a muscle cell in a runner's leg. Present the journey as an annotated diagram or a short illustrated story, naming every structure it passes and explaining what happens at each stage.

What to hand in

- The full journey, in order, with every structure correctly named.
- An explanation of gas exchange at the alveolus, using the word diffusion.
- The word equation for respiration, at the destination.
- One sentence on what happens to the carbon dioxide produced.

How it will be judged

Criterion	Weight
Scientific accuracy and correct order	40%
Quality of explanation at each stage	30%
Use of correct scientific vocabulary	20%
Clarity of presentation	10%

UNIT 1 · CHECK YOUR PROGRESS

1. Name the parts of the respiratory system in the order air passes through. [3]
2. What is the function of the cilia in your airways? [1]
3. a. What is gas exchange? b. Where does it happen? [2]
4. Give three adaptations of an alveolus and explain each. [3]
5. Describe what happens to the diaphragm and ribs when you breathe in. [2]
6. Write the word equation for respiration. [1]
7. Explain the difference between breathing and respiration. [2]
8. a. Which blood cells carry oxygen? b. Name the substance that binds oxygen. [2]
9. A runner's breathing rate rises from 14 to 32 breaths per minute. Explain why, in terms of respiration. [2]

UNIT 1 · EVALUATION

Be honest. Colour the circle that matches how you feel about each outcome: green means confident, amber means nearly there, red means needs more work.

I CAN ...	GREEN	AMBER	RED
name the parts of the respiratory system	○	○	○
explain gas exchange in the alveoli	○	○	○
describe how breathing works	○	○	○
tell breathing from respiration	○	○	○
describe how blood carries oxygen	○	○	○

UNIT 1 · WHAT CAN YOU DO?

✓ WHAT CAN YOU DO?

- ☐ Name the parts of the respiratory system and their functions.
- ☐ Explain gas exchange by diffusion in the alveoli.
- ☐ Describe the action of the diaphragm and ribs.
- ☐ Distinguish breathing from respiration.
- ☐ Describe the parts of blood and how oxygen is carried.

UNIT 1 ROUND-UP

respiration · breathing · trachea · bronchus · bronchiole · alveolus · diaphragm · cilia · diffusion · gas exchange · glucose · mitochondrion · haemoglobin · plasma

GLOSSARY

Unit 1

respiration: the release of energy from food inside cells.

breathing: the movement of air into and out of the lungs.

trachea: the windpipe, carrying air towards the lungs.

alveolus: a tiny air sac where gas exchange happens.

diffusion: movement from a higher to a lower concentration.

diaphragm: the sheet of muscle below the lungs.

cilia: tiny hairs that sweep mucus out of the airways.

mitochondrion: the part of a cell where respiration happens.

haemoglobin: the substance in red blood cells that binds oxygen.

SOURCES & REFERENCES

Where everything came from

All text and diagrams are original to this edition. The factual claims can be checked here. Links were live as of July 2026.

§ UNIT 1 · RESPIRATION

1. **Resting respiratory rate. A healthy resting adult breathes about 12 to 20 times per minute.**
   https://www.lung.org/blog/respiratory-rate-vital-signs
2. **The human respiratory system. Structure and function of the airways, alveoli and diaphragm.**
   https://en.wikipedia.org/wiki/Respiratory_system
3. **Alveoli and surface area. The lungs contain hundreds of millions of alveoli, giving a very large surface area.**
   https://en.wikipedia.org/wiki/Pulmonary_alveolus
4. **Composition of inhaled and exhaled air. The percentages quoted in the gas exchange table.**
   https://en.wikipedia.org/wiki/Breathing

PRIME BOOKS

Science

Student Manual

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INSIDE THIS BOOK

- Biology, chemistry and physics strands
- Practical investigations throughout
- Diagrams, data and vocabulary support
- End-of-unit checkpoints
- Full glossary and answer key

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Ages 12–13 · Lower Secondary

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